Technical Documentation: RP Mulawin FleetOps

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Ang dokumentong ito ang buong paliwanag ng system: paano ito binuo, paano gumagana ang bawat bahagi, saan galing ang data, at paano nagkakabit ang lahat. Pinapalitan nito ang naunang technical doc (v1), na may lumang mga file path bago ang department folders.

1. Overview at tech stack

Ang **Mulawin FleetOps** (RP_Mulawin_FMS_V2) ay web system ng R.P. Mulawin Trucking para sa trips at dispatch, fleet, maintenance at purchasing, trip funds, finance, billing at payroll. Sinusunod nito ang workflow ng FMS PDF (bersyon 2: Scenario A, B at C) at ang mga formula ng Excel files ng kumpanya (Billing P&L at Drivers Payroll).

Bahagi	Gamit	Paliwanag
Server language	PHP 8.2 o mas bago	Walang framework. Plain PHP na may `declare(strict_types=1)`.
Database	MySQL o MariaDB (InnoDB, utf8mb4)	43 tables. Konektado gamit ang PDO na may prepared statements.
Web server	Apache (XAMPP sa local, Hostinger sa live)	Gumagamit ng `.htaccess` para harangin ang mga private folder.
Frontend	Vanilla JavaScript (single-page app)	`core.js` at isang JS file bawat grupo ng pahina. Walang build step.
UI library	Bootstrap 5.3.3 at Bootstrap Icons 1.11.3	Naka-vendor sa `public/assets/vendor` (walang CDN).
Charts	Chart.js 4.4.0	Para sa dashboards at reports. Naka-vendor din.
Timezone	Asia/Manila	Pareho sa PHP at sa MySQL session.

Ang laki ng system

- 11 backend modules, naka-grupo sa 5 department folders.
- 10 roles, 22 na pahina.
- 43 database tables (11 sa `schema.sql`, 32 sa `extension.sql`).
- Mga 7,200 na linya ng PHP at JavaScript (hindi kasama ang vendor libraries).

Mga prinsipyo ng pagkakagawa

1. **Isang pinto lang.** Lahat ng request ay dumadaan sa `public/index.php`.
2. **Modular.** Bawat module ay hiwalay na file. Kapag nasira ang isa, gumagana pa rin ang iba.
3. **Ang server ang nagpapasya.** Ang role check, validation at status rules ay nasa backend, hindi lang sa screen.
4. **Lahat ay nakatala.** Bawat mahalagang galaw ay may record sa `audit_events` (Activity Log).
5. **Formulas na nababago.** Ang mga numero ng computation ay nasa `config/formulas.php` at Formula Settings, hindi naka-hardcode.
6. **Taglish na comments** sa code, para madaling maintindihan ng susunod na developer.

2. Folder structure ayon sa department

Ang mga module ay naka-grupo sa folders ayon sa department. Hindi ito nagbabago ng sidebar; sa code at docs lang ang grupo.

Department	Backend module (PHP)	Screen file (JS)	Mga pahina
Operations	`Operations/Fleet.php`, `Operations/Trips.php`, `Operations/Inspections/Checklist.php`	`operations/fleet.js`, `operations/trips.js`	Fleet Status & Trucks, Drivers Helpers & Staff, Dispatch Board, Trip Records & Waybills, Incidents
Maintenance	`Maintenance/Maintenance.php`, `Maintenance/Purchasing.php`	`maintenance/maintenance.js`, `maintenance/purchasing.js`	Maintenance, Parts Inventory, Purchasing
Accounting	`Accounting/Funds.php`, `Finance.php`, `Billing.php`, `Payroll.php`	`accounting/finance.js`, `billing.js`, `payroll.js`	Trip Funds & Allowances, Finance & Payables, Billing & Collections, Clients Routes & Rates, Payroll, Attendance, Cash Advance & Balances
General	`General/Analytics.php`, `General/Documents.php`	`general/dashboard.js`, `documents.js`, `reports.js`	Dashboard, Documents, Reports & Analytics

Department	Backend module (PHP)	Screen file (JS)	Mga pahina
Administration	`Administration/Admin.php`	`administration/admin.js`	User Accounts, Formula Settings, Activity Log, Backup

Ang Trip Funds ay nasa Accounting sa backend (`Funds.php`), pero ang screen nito ay nasa `accounting/finance.js` dahil Finance ang nagre-release.

Buong folder

```
RP_Mulawin_FMS_V2/
├─ index.php                 # nagre-redirect sa public/
├─ .env                      # DB at app settings (hindi kasama sa git)
├─ public/                   # ito lang ang bukas sa browser
|  ├─ index.php              # HTML shell + JSON API (isang pinto)
|  ├─ install.php            # unang Admin account (burahin pagkatapos)
|  └─ assets/
|     ├─ css/                # layout.css, login.css, app.css
|     ├─ js/core.js          # App, UI, api(), routing, tables, forms
|     ├─ js/modules/<department>/<file>.js
|     └─ vendor/             # Bootstrap, Bootstrap Icons, Chart.js
├─ backend/                  # bawal sa browser (.htaccess)
|  ├─ bootstrap.php          # env, session, db(), helpers, roles, audit()
|  ├─ Actions.php            # router: Module.action at page data
|  ├─ Data.php               # sino ang naka-login, sidebar, notifications
|  ├─ Core/                  # shared na rules ng lahat ng module
|  └─ Modules/<Department>/<Module>.php
├─ config/formulas.php       # default values ng formulas (galing Excel at
PDF)
├─ database/schema.sql       # 11 core tables
├─ database/extension.sql    # 32 tables ng V2 workflow
├─ storage/backups/          # .sql backups
├─ storage/private/          # mga na-upload na documents
├─ scripts/                  # create_user.php, preflight.php
├─ tests/                    # formula_test.php, e2e_workflow.py
└─ docs/                     # guides, traceability, test report
```

Ang Core (shared na rules)

File	Gamit	Gumagamit
`Access.php`	Listahan ng 22 na pahina: label, icon, sidebar section, module, JS file at roles na pwedeng magbukas	Actions, Data
`FleetState.php`	Iisang rule ng status ng truck: Available, Assigned, On Trip, Under Maintenance, Unavailable	Trips, Maintenance, Fleet
`TripFunds.php`	Fund request ng dispatcher (categories, computation, mga status)	Funds, Trips
`Money.php`	Payment details (method, cheque, bank, reference) at voucher number	Funds, Finance, Payroll
`Notify.php`	Notification sa bell. Kapag pumalya, hindi napipigil ang pangunahing transaksyon.	Lahat
`Settings.php`	Formula engine: binabasa ang `config/formulas.php` at ang mga binago sa Formula Settings	Trips, Fleet, Maintenance, Purchasing, Billing, Payroll, Documents, Analytics, Admin (at Funds sa TripFunds)

3. Architecture at request lifecycle

Bawat request ay dumadaan sa iisang pinto bago umabot sa database



request lifecycle · browser hanggang database

Pagbalik, sumusunod ang sagot sa parehong daan pataas: ang module ay nagbabalik ng PHP array, ginagawang JSON ng `index.php` (`{"ok": true, "data": ...}` o `{"ok": false, "error": "..."}`), at ipinapakita ng screen.

A. Pagbukas ng isang pahina (read)

1. Pinindot ng user ang pahina sa sidebar. Nagbabago ang URL hash, halimbawa `#trips` o `#payroll?period=3`.
2. `App.route()` sa `core.js` ang humahanap ng page definition (`App.page('trips', ...)`).
3. `App.data('trips', query)` ang tumatawag sa `GET index.php?api=data&page=trips`.
4. Sa `index.php`, ang `case 'data'` ay tumatawag sa `Actions::pageData()`.

5. `currentUser()` ang tumitingin ng session, idle timeout (default 60 minuto) at kung active pa ang user.
6. `Access::canOpen()` ang tumitingin kung kasama ang role sa listahan ng pahina. Kung hindi: 403.
7. `Actions::load()` ang nagre-require ng module file (halimbawa `Modules/Operations/Trips.php`).
8. `Trips::read($u, 'trips', $query)` ang nagbabasa sa database gamit ang `readSections()`. Kapag pumalya ang isang bahagi, walang laman lang ang bahaging iyon at may maliit na babala; tuloy ang iba.
9. Ibinabalik ang data bilang JSON at ang `render(ctx)` ng page ang gumuguhit ng tables, cards at charts.

B. Pag-save o pag-approve (action)

1. Pinindot ng user ang button, halimbawa **Confirm Release**. Ang `UI.formModal()` ang nagpapakita ng form.
2. `App.api('Funds.release', data)` ang nagpapadala ng `POST index.php?api=Funds.release` na may header na `X-CSRF-Token`.
3. Sa `index.php`, ang `default` case ay nagche-check ng POST at CSRF, tapos `Actions::run()`.
4. Sinusuri ng `Actions::run()` ang pangalan (`Module.action`), ang user, at kung kailangan pang palitan ang password.
5. `Funds::action()` ang pumipili ng private function gamit ang `match`.
6. Sa loob: `requireRole()` para sa role, mga field helper (`idField`, `moneyField`, `dateField`) para sa input, at `must()` para sa business rules.
7. Lahat ng pagsulat ay nasa `transaction()`. Kapag may error kahit saan, walang nase-save (rollback).
8. Sa parehong transaksyon: `audit()` para sa Activity Log at `Notify::roles()` o `Notify::user()` para sa bell.
9. Ibinabalik ang `message` (at `warning` kung meron). Ang `UI.result()` ang nagpapakita ng toast at nire-reload ang pahina.

C. Paano hinahawakan ang errors

Uri	Galing sa	HTTP code	Makikita ng user
Business rule	`must()` o `BusinessRuleException`	400, 403, 404, 409	Ang mismong mensahe, halimbawa "The fund is not Ready for Release yet"
Hindi naka-login	`currentUser()`	401	Bumabalik sa login screen
Sirang module	`Actions::load()`	503	"The X module is temporarily unavailable. Other modules are still working."
Database error	`PDOException`	500	"This record already exists (duplicate)." o pangkalahatang mensahe; detalye ay nasa server error log
Iba pang error	`Throwable`	500	"Unexpected error. See the server error log."

4. Frontend: paano gumagana ang screen

Ang screen ay **single-page app**: isang HTML page lang ang nilo-load (`public/index.php` na walang `?api=`), at ang JavaScript ang nagpapalit ng laman. Walang build tool; diretso ang mga `.js` file.

A. Pagkakasunod ng pag-load

1. Nilo-load ng HTML shell ang Bootstrap, Chart.js, `core.js` , at ang 11 na screen files (`$jsModules` sa `index.php`). May `?v=<filemtime>` ang bawat file para laging bago kapag na-update.
2. Bawat screen file ay nagre-register ng mga pahina: `App.page('payroll', { title, subtitle, render(ctx) })` .
3. `App.start()` ang tumatawag sa `?api=whoami` . Kung hindi naka-login, login screen. Kung naka-login, `?api=me` .

4. Ang `?api=me` (`backend/Data.php`) ang nagbabalik ng user, CSRF token, sidebar (`Access::navFor`), notifications, pangalan at address ng kumpanya, truck types at role labels.
5. Kung `must_change_password`, ang change password screen muna. Kung hindi, `App.shell()` ang gumuguhit ng sidebar at header.
6. Bawat 60 segundo, `refreshBell()` ang kumukuha ulit ng notifications.

B. Ang `core.js`

Bahagi	Gamit
`App.request`, `App.api`, `App.data`, `App.upload`	Lahat ng tawag sa server. Kapag 401, balik sa login.
`App.route`, `App.go`	Router gamit ang URL hash (`#page?x=y`). Pahinang wala sa sidebar ng role ay balik sa Dashboard.
`ctx` ng bawat page	`page`, `query`, `el`, `tools`, `me`, `data`, `reload()`, `go()`, `setQuery()`
`UI.table`, `UI.activate`	Tables na may Search, bilang ng records, **CSV**, Prev at Next (25 rows bawat page), at action buttons bawat linya
`UI.formModal`, `UI.confirmAction`, `UI.modal`	Mga form at confirm. Pagka-save, toast at reload ng pahina.
`UI.field`, `UI.fields`, `UI.formData`	Mga input: text, number, money, date, select, textarea, checkbox, file at iba pa
`UI.blocks`, `UI.cards`, `UI.chartHtml`, `UI.drawChart`	Dashboard cards (may grupo kapag may group ang card), charts (maxCharts, ang iba ay nasa Show more) at tables (higit sa isa = iisang card na may pills; walang laman = isang linya) mula sa iisang spec ng backend
`UI.tabs`	Tabs sa loob ng pahina (naaalala ang huling tab). Ang UI.groupTabs ay tabs na may pills sa loob, halimbawa sa Finance at Formula Settings.
`UI.fundForm`	Form ng fund request (categories at total)

Bahagi	Gamit
`UI.print`	Pag-print ng payslips, vouchers at forms
`UI.csv`	Excel-friendly na CSV (may BOM, at ligtas sa formula injection)
`UI.money`, `UI.date`, `UI.badge`, `UI.esc`	Format ng pera (₱), petsa, status badge, at escape ng text laban sa XSS

C. Ang isang page, halimbawa

```
App.page('audit_log', {
  title: 'Activity Log',
  render(ctx) {
    const d = ctx.data;                    // galing sa Admin::read(...,
'audit_log')
    ctx.el.innerHTML = UI.table([...columns], d.events, { id: 'alog' });
    UI.activate(ctx.el);                   // binubuhay ang search, CSV at
buttons
  },
});
```

D. Hiwalay ang bawat page

- Kapag hindi na-load ang isang screen file, ang pahina lang na iyon ang may mensaheng "This module's screen … did not load. Other modules still work."
- Kapag nag-error ang data o render ng isang pahina, "Only this page is affected." Gumagana pa rin ang sidebar at ibang pahina.
- Kapag may `_errors` ang data (isang bahagi ang pumalya), dilaw na babala lang sa itaas ng pahina.

E. CSS

`layout.css` (sidebar, header, cards), `login.css` (login screen) at `app.css` (tables, badges, forms, print). Kulay: green na sidebar at gold na accent.

F. Navigation (polish, 2026-11-01)

Para hindi mahaba at hindi nakaka-overwhelm ang mga page. Walang binago sa data, workflow o database.

Ano	Saan sa code	Paano gumagana
Accordion na sidebar	`App.openSection()` sa `core.js`	Isang section lang ang bukas; ang `route()` ang nagbubukas ng section ng current page
Nakadikit na header at Back to top	`.page-header-sticky`, `#toTop`, `window.onscroll`	Nakadikit ang title at buttons; nawawala ang subtitle habang naka-scroll; lumalabas ang Back to top pagkalipas ng 400px
Compact cards sa work pages	`#pageBody.work-page` sa `app.css`	Lahat ng page maliban sa Dashboard at Reports
Pagination	`UI.drawTable` (`t.page`, `t.limit`)	25 rows bawat page; Search at CSV ay sakop ang lahat ng rows
Reports tabs at More filters	`general/reports.js`	Tab bawat section ng `Analytics::reports`; ang ibang filters ay nakatago hangga't walang gamit
Dashboard	`general/dashboard.js` (`maxCharts: 2`), `Analytics::group()`	Mga card ng Admin at Ops Head ay may grupo (Trips, Fleet, Pera)
Formula Settings	`administration/admin.js`	Tabs: Values (pill bawat group, editable muna, View only sa huli), Computation lists, Example
Finance & Payables	`accounting/finance.js`	Tabs: Gastos at Bayarin, Empleyado, Plano

Resulta (pinakamahabang page): Reports 6,949px hanggang 1,454px, Formula Settings 4,806px hanggang 900px, Admin Dashboard 2,706px hanggang 1,557px. Wala nang page na lampas 1,700px sa test data.

5. API reference

Internal API ang system: ginagamit lang ng sariling screen, hindi public. Lahat ay nasa `public/index.php?api=...`, JSON ang sagot, at kailangan ng naka-login na session (maliban sa `whoami` at `login`).

A. Mga endpoint

Endpoint	Method	Ginagawa	Sumasagot
`?api=whoami`	GET	Kung naka-login at ang CSRF token	`{ok, signedIn, csrf}`
`?api=login`	POST (form + csrf)	Login. 5 maling subok sa loob ng 15 minuto ay haharangin.	`{ok}`
`?api=logout`	POST	Logout at sira ang session	`{ok}`
`?api=me`	GET	User, sidebar, notifications, company info, roles	`Data::me()`
`?api=data&page=<id>`	GET	Data ng isang pahina (kasama ang filters sa query string)	`<Module>::read()`
`?api=<Module>.<action>`	POST JSON + `X-CSRF-Token`	Pag-save, approve, release at iba pa	`<Module>::action()`
`?api=document_upload`	POST multipart	Upload ng isa o maraming file	`Documents::upload()`
`?api=document_download&id=<n>`	GET	Download o view (`&inline=1`) ng document	ang file

Endpoint	Method	Ginagawa	Sumasagot
`?api=backup_download&file=<name>`	GET (Admin lang)	Download ng .sql backup	ang file

Halimbawa ng tawag mula sa screen:

```
await App.api('Funds.release', { fund_id: 6, payment_method: 'Cash',
released_on: '2026-11-01' });
// POST index.php?api=Funds.release  →  {"ok":true,"data":
{"fund_id":6,"message":"Fund released and disbursement recorded."}}
```

B. Lahat ng actions

Ang "Tables" ay ang mga sinusulatan ng action (bukod sa `audit_events` at `notifications`, na halos lahat ay sinusulatan). "Dispatcher (sariling team)" = Car Carrier o Container / Wing Van Dispatcher, at dapat kanya ang trip.

Operations: `Trips` (Modules/Operations/Trips.php)

Action	Sino	Ginagawa	Tables
`Trips.create`	Operations Head, Dispatcher	Bagong trip (Scenario A) o delivery request (Scenario B)	trips, trip_events
`Trips.review`	Operations Head	Approve o reject ng delivery request	trips, trip_events
`Trips.auto_assign`	Operations Head, Dispatcher	Subukang mag-auto-assign ulit ng truck at crew	trips, vehicles, trip_events
`Trips.reassign`	Operations Head	Manual na palit ng truck o crew, may dahilan	trips, vehicles
`Trips.details`	Operations Head, Dispatcher	I-edit ang detalye ng trip	trips

Action	Sino	Ginagawa	Tables
`Trips.checklist_start`	Dispatcher (sariling team)	Simulan ang Departure o Arrival checklist	inspections
`Trips.checklist_departure`	Dispatcher (sariling team)	I-save ang pre-dispatch checklist → Ready for Dispatch	inspections, trips
`Trips.dispatch`	Dispatcher (sariling team)	Confirm Dispatch (kailangan Released ang fund)	trips, fund_requests, vehicles, personnel, vehicle_status_log
`Trips.progress`	Dispatcher (sariling team)	Trip update o delay	trip_updates
`Trips.incident`	Dispatcher (sariling team)	Report incident → On Hold, gagawa ng maintenance job	incidents, maintenance_jobs, trips
`Trips.arrival`	Dispatcher (sariling team)	Arrival checklist at odometer → Returned	inspections, trips, vehicles, maintenance_jobs
`Trips.complete`	Dispatcher (sariling team)	Mark as Completed	trips
`Trips.resume`	Dispatcher (sariling team)	Ituloy ang trip pagkatapos ma-clear ang repair	incidents, trips
`Trips.reschedule`	Dispatcher (sariling team)	Bagong schedule ng trip na For Reschedule	trips
`Trips.cancel`	Operations Head	Cancel ng trip (at ng fund nito)	trips, fund_requests

Action	Sino	Ginagawa	Tables
`Trips.waybill_add`	Dispatcher, Admin, Billing	I-record ang waybill	waybills

Operations: `Fleet` (Modules/Operations/Fleet.php)

Action	Sino	Ginagawa	Tables
`Fleet.vehicle_save`	Admin	Bago o edit ng truck	vehicles
`Fleet.vehicle_status`	Admin, Motorpool	Palit ng status ng truck	vehicles, vehicle_status_log
`Fleet.vehicle_active`	Admin	Active o inactive na truck	vehicles
`Fleet.person_save`	Admin	Bago o edit ng driver, helper o staff	personnel
`Fleet.person_pay`	Admin, Payroll	Pay info (daily rate, frequency, group)	personnel
`Fleet.person_availability`	Admin, Operations Head	Available o unavailable na tao	personnel

Maintenance: `Maintenance` at `Purchasing`

Action	Sino	Ginagawa	Tables
`Maintenance.job_add`	Motorpool	Manual na maintenance job	maintenance_jobs, vehicles

Action	Sino	Ginagawa	Tables
`Maintenance.assess`	Motorpool	Assess ng incident: minor (repair on site) o major (For Reschedule)	maintenance_jobs, incidents, trips, trip_events
`Maintenance.job_update`	Motorpool	Status, mechanic, expected date	maintenance_jobs
`Maintenance.job_complete`	Motorpool	Tapos ang repair → truck Available, PM reset	maintenance_jobs, incidents, maintenance_schedules, vehicles
`Maintenance.part_request`	Motorpool	Humingi ng parts sa Purchasing	part_requests, maintenance_jobs
`Maintenance.part_use`	Motorpool	Gamit na parts sa job	maintenance_parts_used
`Maintenance.cost_add`	Motorpool	Labor o ibang gastos	maintenance_costs, expenses
`Maintenance.schedule_add` / `schedule_start` / `schedule_cancel`	Motorpool	Preventive maintenance schedule	maintenance_schedules
`Purchasing.supplier_save`	Purchasing	Supplier	suppliers
`Purchasing.part_save`	Purchasing	Part sa inventory	parts
`Purchasing.adjust`	Purchasing	Adjust stock	parts, stock_movements
`Purchasing.issue`	Purchasing	Release parts sa part request	parts, stock_movements, part_requests, maintenance_jobs
`Purchasing.not_available`	Purchasing	Walang stock ng hiniling	part_requests

Action	Sino	Ginagawa	Tables
`Purchasing.pr_create`	Purchasing	Purchase request (Finance kung ≤ ₱10,000, Admin kung lampas)	purchase_requests
`Purchasing.decide`	Admin o Finance	Approve (may budget check) o reject	purchase_requests, budgets
`Purchasing.po_create`	Purchasing	Purchase order	purchase_orders, purchase_requests
`Purchasing.receive`	Purchasing	Stock-in at gagawa ng payable	purchase_orders, parts, stock_movements, accounts_payable, expenses

Accounting: `Funds`, `Finance`, `Billing`, `Payroll`

Action	Sino	Ginagawa	Tables
`Funds.request`	Dispatcher (sariling team)	Fund request para sa Pre-Assigned trip	fund_requests
`Funds.revise`	Dispatcher (sariling team)	Ayusin at ipasa ulit ang na-reject	fund_requests
`Funds.approve` / `Funds.reject`	Operations Head	Approve → Ready for Release, o reject na may dahilan	fund_requests
`Funds.release`	Finance	Confirm Release	fund_requests, payment_vouchers, disbursements, expenses

Action	Sino	Ginagawa	Tables
`Funds.liquidate`	Finance	Liquidation ng nagastos	fund_requests, expenses
`Finance.expense_add`	Finance	Gastos	expenses, disbursements
`Finance.payable_add` / `payable_pay`	Finance	Payable at bayad nito	accounts_payable, disbursements
`Finance.budget_set`	Finance	Budget (halimbawa Purchasing)	budgets
`Finance.cash_advance`	Finance	Cash advance sa empleyado	disbursements, employee_ledgers, ledger_entries
`Finance.payroll_disburse`	Finance	Bayaran ang payroll na na-submit	payroll_periods, disbursements, expenses
`Billing.client_save` / `rate_save`	Admin, Billing	Client at client rates	clients, client_rates
`Billing.route_save`	Admin, Operations Head, Billing	Route (km at RFID/toll)	routes
`Billing.invoice_create`	Billing	Invoice mula sa completed trips	billings, billing_items
`Billing.invoice_cancel`	Billing	Cancel (kung walang bayad pa)	billings, billing_items
`Billing.collection_add`	Billing	Bayad ng client	collections, billings
`Payroll.rate_save`	Payroll	Trip pay rate	trip_pay_rates
`Payroll.period_create` / `period_compute`	Payroll	Bagong payroll at recompute	payroll_periods, payroll_lines
`Payroll.line_update` / `line_add` / `line_remove`	Payroll	Ayusin ang linya ng empleyado	payroll_lines

Action	Sino	Ginagawa	Tables
`Payroll.finalize` / `reopen` / `submit`	Payroll	Finalize (ibabawas ang hulog), reopen, submit sa Finance	payroll_periods, ledger_entries, employee_ledgers
`Payroll.attendance_save`	Payroll	Attendance	attendance
`Payroll.ledger_add` / `ledger_payment` / `ledger_installment`	Payroll	Cash advance, uniform, damages, cash bond	employee_ledgers, ledger_entries

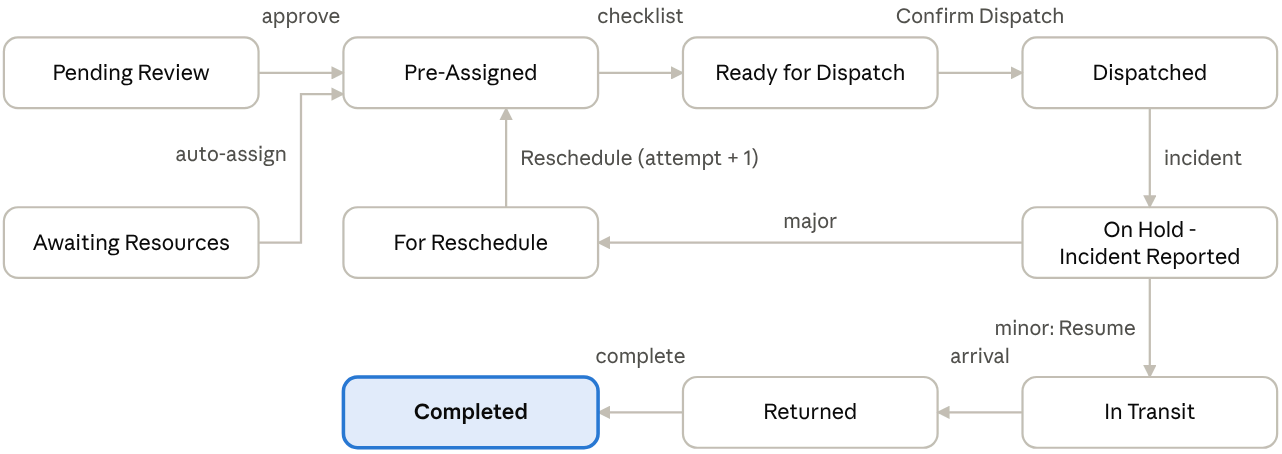
Administration: **`Admin`** (General: `Analytics` ay read lang; `Documents` ay sa upload/download endpoint)

Action	Sino	Ginagawa	Tables
`Admin.user_save` / `user_toggle` / `user_password`	Admin	Account, activate/deactivate, reset password	users
`Admin.change_password`	Lahat (sarili)	Palit ng sariling password	users
`Admin.setting_save` / `setting_reset`	Ayon sa group (Admin, Finance, Payroll)	Isang formula value	app_settings
`Admin.items_save` / `items_reset`	Ayon sa listahan	Computation lists	formula_items
`Admin.backup_create`	Admin	Gumawa ng .sql backup	(file sa storage/backups)
`Admin.notifications_read`	Lahat	Markahan na nabasa	notifications

6. Paano gumagana ang bawat module sa backend

A. Operations: Trips at Fleet

Ang trip ay dumadaan sa checklist at released fund bago ma-dispatch



trip status flow · Scenario A, B at C

Ang Scenario A ay diretso sa auto-assign (Pre-Assigned o Awaiting Resources). Ang Scenario B ay dumadaan muna sa Pending Review. Ang arrival ay galing sa Dispatched o In Transit. Ang Rejected (review) at Cancelled (Operations Head, bago umalis) ay mga dulo na hindi na tumutuloy.

Auto-assign (`Trips::autoAssign`). Kapag na-confirm ang trip, naghahanap ang system ng truck at crew:

1. Truck: active, tamang truck type, walang bukas na maintenance job, at walang ibang trip sa parehong araw. Kung ngayong araw ang trip, Available lang; kung sa susunod, pwede rin ang Assigned o On Trip.
2. Pagkakasunod: unahin ang hindi pa aabot sa preventive maintenance (10,000 km), tapos ang may pinakakaunting trip sa huling 30 araw (fair rotation), tapos ang pinakamababang km mula sa huling PM.
3. Driver: may valid na license, libre sa araw na iyon, at pinakakaunting trip. Kung kasama sa listahan ang regular driver ng truck, siya ang pipiliin.
4. Car Carrier: dalawang driver. Container at Wing Van: driver at helper.
5. Kapag kumpleto: `Pre-Assigned`, truck ay `Assigned`. Kapag kulang: `Awaiting Resources` na may dahilan.

6. Naka-lock ang truck row (`FOR UPDATE`) para hindi makuha ng dalawang dispatcher ang iisang truck.

Confirm Dispatch (`Trips::dispatchBlockers`). Enabled lang kapag ang trip ay `Ready for Dispatch` at ang fund ay `Released` . Sa redispatch (attempt 2 pataas), optional ang fund. Pagka-dispatch: truck at crew ay `On Trip` , fund ay `Completed` .

Arrival. Kailangang mas mataas ang arrival odometer kaysa departure. Idinadagdag ang km sa truck; kapag umabot sa 10,000 km mula sa huling PM, gagawa ng preventive maintenance schedule (at kung naka-on ang auto hold, Under Maintenance ang truck).

Bawat pagbabago ng status ay may record sa `trip_events` at `audit_events` . Ang `FleetState` ang iisang lugar na nagbabago ng status ng truck at nagtatala sa `vehicle_status_log` .

B. Accounting: Trip Funds

1. Dispatcher: `Funds.request` gamit ang preset categories (Forklift Operator 100, Checker 100, Passway 200, Driver Meal Allowance 300, Helper Meal Allowance 200, mula sa Formula Settings). Status: `Pending Approval` .
2. Operations Head: approve → `Ready for Release` , o reject na may dahilan → `Rejected` (pwedeng i-revise at ipasa ulit).
3. Finance: `Funds.release` → `Released` . Gagawa ng payment voucher, disbursement at expense (category Trip Allowance).
4. Dispatch → `Completed` . Kapag na-cancel ang trip → `Cancelled` .
5. Finance: liquidation. Ang expense ay papalitan ng aktuwal na nagastos at itatala ang ibinalik.

C. Maintenance at Purchasing

1. Incident mula sa Operations → `incidents` (Reported) at `maintenance_jobs` (source Incident).
2. `Maintenance.assess` : **Minor** → incident Under Repair, mechanic sa lugar. **Major** → truck Under Maintenance, trip at incident For Reschedule.
3. `Maintenance.job_complete` : incident `Repaired - Cleared to Continue` (minor) o `Resolved` (major), truck balik sa Available sa pamamagitan ng `FleetState::reconcile` , at reset ng PM km kung preventive.
4. Parts: `part_request` → Purchasing `issue` (bawas stock) o `not_available` → `pr_create` .

5. Purchase request: approver ay Finance kung ≤ ₱10,000, Admin kung lampas. Sa approve, sinusuri ang active na Purchasing budget; kapag kulang, kusang `Rejected - Insufficient Budget`.
6. `po_create` → `receive`: dagdag stock, `stock_movements`, `accounts_payable` at expense.

D. Accounting: Finance, Billing at Payroll

- **Finance:** expenses, payables at bayad, budgets, cash advance, at disbursement ng payroll. Lahat ng labas ng pera ay dumadaan sa `Money::paymentFields` at `Money::voucher`, kaya pare-pareho ang payment details at may voucher number.
- **Billing:** invoice mula sa completed trips. Bawat linya: rate × quantity (Per Trip, Per Unit o Per KM). Net = gross − discount. Kapag walang waybill ang trip, nagagawa pa rin pero may babala. `collection_add` ang nagbabawas ng balance ng invoice.
- **Payroll:** `period_create` ang kumukuha ng tao sa payroll group, attendance at completed trips na may waybill, tapos kinukuwenta ang bawat linya gamit ang `Settings` (tingnan ang section 9). `finalize` ang nagbabawas ng hulog sa `employee_ledgers`; `reopen` ang nagbabalik nito; `submit` ang nagpapasa sa Finance.

E. General: Analytics at Documents

- **Analytics** ay read lang. Iba ang dashboard bawat role (Management, Operations, Dispatcher, Maintenance, Purchasing, Finance, Billing, Payroll, Audit). Iisang format ang ibinabalik (`title`, `alerts`, `cards`, `charts`, `tables`) na ginuguhit ng `UI.blocks`. Tuwing nilo-load ang dashboard, tumatakbo ang `Notify::dailyReminders` (expiring documents at licenses, low stock, maintenance due), isang beses bawat araw bawat record.
- **Documents:** upload sa `storage/private` (random na pangalan, hindi ang orihinal; PDF, JPG, PNG o WebP lang, hanggang 10 MB bawat file), record sa `documents`, at may rules kung aling role ang pwedeng mag-upload o tumingin ng bawat uri. Download ay dumadaan lang sa `?api=document_download`.

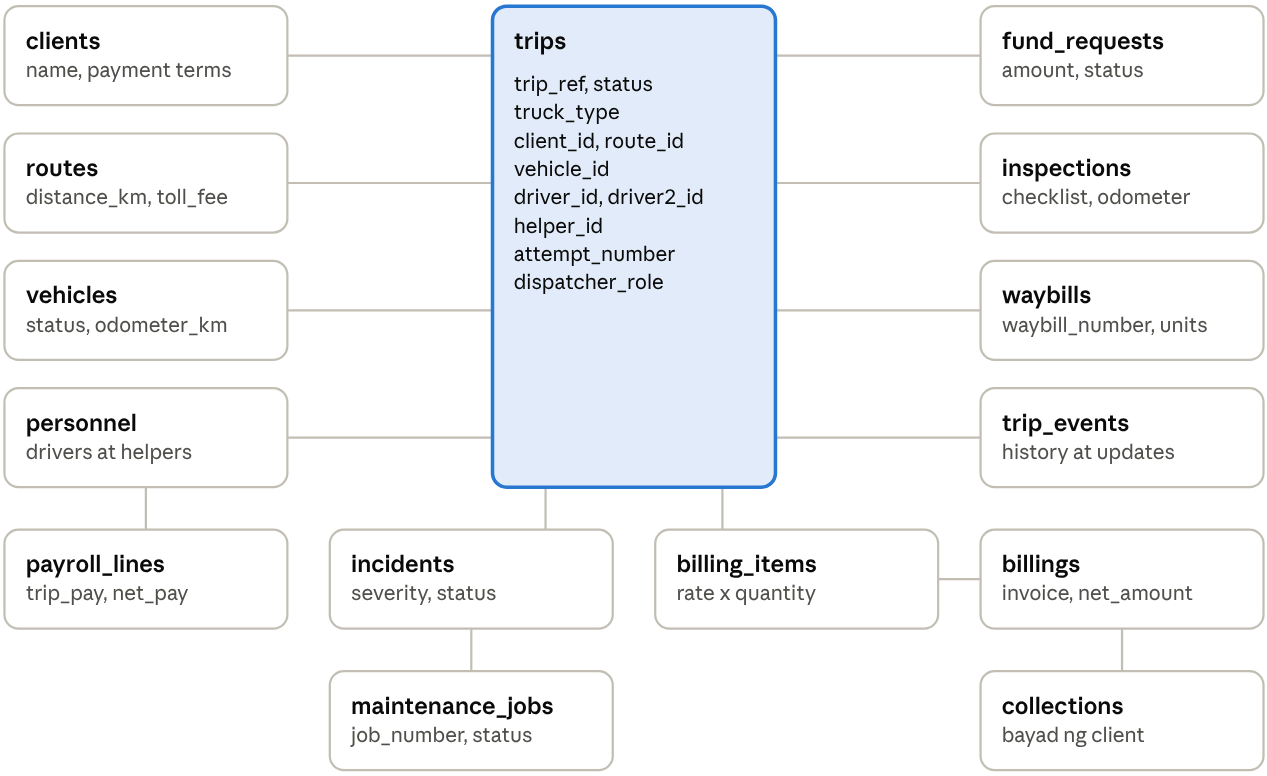
F. Administration

Users, Formula Settings (`app_settings` at `formula_items`), Activity Log (`audit_events`) at Backup (sinusulat ng PHP ang buong database bilang .sql sa `storage/backups`).

7. Database

MySQL o MariaDB, InnoDB engine, `utf8mb4`. 43 tables: 11 sa `database/schema.sql` (core) at 32 sa `database/extension.sql` (V2 workflow). I-import nang magkasunod. May 68 na foreign keys, kaya hindi pwedeng mag-save ng record na tumuturo sa wala.

Ang trips ang sentro: halos lahat ng record ay nakakabit sa isang trip



pangunahing relationships · bawat linya ay foreign key

Sa kaliwa ang mga "master" na tinuturo ng trip; sa kanan ang mga record na ginagawa habang tumatakbo ang trip. Ang `payroll_lines` ay nakakabit sa `personnel`; ang `maintenance_jobs` sa `incidents` (at sa `vehicles` at `trips`); ang `billing_items` sa `trips` at `billings`.

A. Lahat ng tables ayon sa area

Area	Table	Laman at mahahalagang columns	Nakakabit sa (FK)
Access	`users`	username, password_hash, full_name, role, shift, active, must_change_password, last_login_at	
Access	`audit_events`	Activity Log: actor_id, area, record_ref, action, details, created_at	users
Access	`notifications`	Bell: user_id, category, message, page, dedupe_key, is_read	users
Settings	`app_settings`	Binagong formula values: setting_key, setting_value	users
Settings	`formula_items`	Computation lists: scope, name, calc, amount, divisor, only_if_attended, ledger, active	users
Fleet	`vehicles`	plate_number, truck_type, OR/CR, registration at insurance, status, odometer_km, km_since_pm, assigned driver at helper	personnel
Fleet	`vehicle_status_log`	Kasaysayan ng status ng truck: from_status, to_status, reason	vehicles, users
Fleet	`personnel`	Drivers, helpers, staff: person_type, employment_status, license_expiry, availability, payroll_group, pay_frequency, daily_rate	
Operations	`trips`	Ang trip: trip_ref, intake_source, truck_type, dispatcher_role, route, schedule, container/seal/shipment, client_rate, truck at crew, status, attempt_number	clients, routes, vehicles, personnel, users
Operations	`trip_events`	Bawat palit ng status: from_status, to_status, notes	trips, users

Area	Table	Laman at mahahalagang columns	Nakakabit sa (FK)
Operations	`trip_updates`	Progress at delay: update_type, delay_cause, note	trips, users
Operations	`inspections`	Departure at Arrival checklist: items_json, odometer_km, fuel_level, has_not_ok	trips, vehicles, users
Operations	`incidents`	Incident: severity (Minor/Major), details, problem, location, status	trips, vehicles, users
Operations	`waybills`	waybill_number, client_waybill_number, dr_number, units, submitted_on	trips, users
Maintenance	`maintenance_jobs`	job_number, source (Manual, Incident, Preventive), maintenance_type, status, expected_completion	vehicles, incidents, trips
Maintenance	`maintenance_schedules`	Preventive maintenance: due_on, due_odometer_km, status	vehicles, maintenance_jobs
Maintenance	`maintenance_costs`	Labor at iba pang gastos ng job	maintenance_jobs
Maintenance	`maintenance_parts_used`	Parts na nagamit sa job, quantity, unit_cost	maintenance_jobs, part_requests, parts
Purchasing	`parts`	part_number, description, unit_cost, reorder_level, quantity_on_hand	suppliers
Purchasing	`stock_movements`	Bawat galaw ng stock: movement_type, quantity_delta, reference	parts
Purchasing	`part_requests`	Hiling ng Motorpool: requested_qty, status	vehicles, maintenance_jobs, parts
Purchasing	`purchase_requests`	pr_number, quantity, amount, approver_role, status, budget_id	parts, part_requests, suppliers
Purchasing	`purchase_orders`	po_number, ordered_on, expected_on, received_on, warranty_until	purchase_requests, suppliers
Purchasing	`suppliers`	Supplier details	

Area	Table	Laman at mahahalagang columns	Nakakabit sa (FK)
Funds	`fund_requests`	amount, breakdown_json, status, approve, release at liquidation details	trips
Finance	`expenses`	category, amount, incurred_on, source at source_id (saan galing)	trips, vehicles
Finance	`disbursements`	Lahat ng labas ng pera: source_type, payment_method, cheque, bank, reference, amount	payment_vouchers, users
Finance	`payment_vouchers`	voucher_number, reference_type, payee, amount, status	users
Finance	`accounts_payable`	Utang sa supplier: amount, paid_amount, due_on, status	suppliers
Finance	`budgets`	category, budget_amount, used_amount, period, status	users
Billing	`clients`	name, company_name, contact, payment_terms_days	
Billing	`routes`	origin, destination, distance_km, toll_fee	
Billing	`client_rates`	truck_type, rate_basis, amount, effective_on	clients, routes
Billing	`billings`	invoice_number, invoice_date, due_date, gross_amount, discount, net_amount, status	clients
Billing	`billing_items`	Linya ng invoice: rate_basis, quantity, rate, amount	billings, trips
Billing	`collections`	Bayad: amount, payment_date, payment_method, reference	billings
Payroll	`trip_pay_rates`	driver_rate at helper_rate bawat truck type at route o destination	routes
Payroll	`attendance`	work_date, status (Present, Absent, VL, SIL, Rest Day)	personnel

Area	Table	Laman at mahahalagang columns	Nakakabit sa (FK)
Payroll	`payroll_periods`	period_name, payroll_group, date_from, date_to, status, formula_snapshot	
Payroll	`payroll_lines`	daily_rate, days_worked, trips_count, regular_wage, trip_pay, earnings_json, deductions_json, gross_pay, net_pay	payroll_periods, personnel
Payroll	`employee_ledgers`	Cash Advance, Uniform, Damages, Cash Bond: total_amount, installment, balance, status	personnel
Payroll	`ledger_entries`	Bawat bawas o bayad sa ledger	employee_ledgers
Records	`documents`	category, entity_type, entity_id, original_filename, storage_name, expires_on	users

B. Mga patakaran sa data

- **Pera** ay `DECIMAL`, hindi float, at ipinapasok bilang string mula sa `moneyField()`.
- **Mga status** ay plain text na sinusuri ng code (halimbawa `Trips::STATUSES`), para madaling basahin sa phpMyAdmin.
- **`attempt_number`** sa trips, fund_requests, inspections at incidents ang naghihiwalay ng unang biyahe at ng redispatch (Second Attempt).
- **`source` at `source_id`** sa expenses, accounts_payable at employee_ledgers ang nagsasabi kung saan galing ang record (halimbawa "Fund Release" at ang id ng fund request).
- **`formula_snapshot`** sa payroll_periods ang nagtatago ng formula values na ginamit (sine-save pagka-Finalize), kaya hindi nagbabago ang lumang payroll kahit baguhin ang Formula Settings.
- **Reference numbers** (TRP para sa trip, JOB, PR, PO at PV para sa voucher) ay ginagawa ng `refNumber()` mula sa prefix at id. Ang invoice number ay tina-type ng Billing at sinusuring hindi pa nagamit.

8. Saan kinukuha ng bawat pahina ang data

Bawat pahina ay may iisang `read()` sa module nito. Ang `Access.php` ang nagsasabi kung aling module ang may-ari ng pahina. Ang listahan sa ibaba ay ang mga table na binabasa (SELECT at JOIN).

Pahina (sidebar)	Page id	Backend	Mga table na binabasa
Dashboard	`dashboard`	`Analytics::read` (iba bawat role)	Ayon sa role: trips, vehicles, personnel, maintenance_jobs, fund_requests, billings, collections, expenses, disbursements, payroll_periods, payroll_lines, parts, purchase_requests, documents, audit_events. Sumusulat din ng notifications (daily reminders).
Fleet Status & Trucks	`fleet`	`Fleet::read`	vehicles, personnel (regular crew), trips (aktibong trip), maintenance_jobs, vehicle_status_log, users
Drivers, Helpers & Staff	`crew`	`Fleet::read`	personnel, trips, vehicles, clients
Dispatch Board	`trips`	`Trips::read`	trips, clients, routes, vehicles, personnel, fund_requests, inspections, trip_updates, trip_events, users. Checklist items mula sa `Inspections/Checklist.php`; fund categories mula sa formula_items.

Pahina (sidebar)	Page id	Backend	Mga table na binabasa
Trip Records & Waybills	`trip_records`	`Trips::read`	trips, waybills, trip_updates, trip_events, inspections, fund_requests, billing_items, clients, vehicles, personnel, users. Billing at Payroll: Completed lang. Dispatcher: sariling team lang.
Incidents	`incidents`	`Trips::read`	incidents, trips, vehicles, maintenance_jobs, clients, users
Maintenance	`maintenance`	`Maintenance::read`	maintenance_jobs, incidents, trips, vehicles, maintenance_schedules, maintenance_costs, maintenance_parts_used, part_requests, parts, users
Parts Inventory	`inventory`	`Purchasing::read`	parts, suppliers, part_requests, maintenance_jobs, vehicles, stock_movements, users
Purchasing	`purchasing`	`Purchasing::read`	purchase_requests, purchase_orders, parts, part_requests, suppliers, budgets, vehicles, users
Trip Funds & Allowances	`funds`	`Funds::read`	fund_requests, trips, clients, vehicles, users (at formula_items para sa categories)
Finance & Payables	`finance`	`Finance::read`	expenses, accounts_payable, suppliers, payment_vouchers, disbursements, budgets, payroll_periods, payroll_lines, employee_ledgers, personnel, vehicles, trips, users

Pahina (sidebar)	Page id	Backend	Mga table na binabasa
Billing & Collections	`billing`	`Billing::read`	trips (completed, hindi pa naka-bill), waybills, vehicles, billings, billing_items, collections, clients, users
Clients, Routes & Rates	`clients`	`Billing::read`	clients, routes, client_rates, billings, collections, trips
Payroll	`payroll`	`Payroll::read`	payroll_periods, payroll_lines, personnel, trip_pay_rates, routes (at app_settings at formula_items para sa computation)
Attendance	`attendance`	`Payroll::read`	personnel, attendance
Cash Advance & Balances	`balances`	`Payroll::read`	employee_ledgers, ledger_entries, personnel
Documents	`documents`	`Documents::read`	documents, users (at ang record na kinakabitan para sa link)
Reports & Analytics	`reports`	`Analytics::read`	Mga section ayon sa role: Trips, Fleet, Money, Payroll at Stock. May filters: petsa, grupo (day, week, month), client, truck, driver, dispatcher, truck type, status.
Formula Settings	`settings`	`Admin::read`	app_settings, formula_items, at default sa `config/formulas.php`
User Accounts	`users`	`Admin::read`	users

Pahina (sidebar)	Page id	Backend	Mga table na binabasa
Activity Log	`audit_log`	`Admin::read`	audit_events, users (Audit: Finance, Billing, Purchasing, Payroll, Documents, Operations lang)
Backup	`backup`	`Admin::read`	Mga file sa `storage/backups` (hindi table)

Halimbawa: saan galing ang "Outstanding receivables" sa Management Dashboard

1. Ang screen ay tumatawag ng `?api=data&page=dashboard`.
2. Ang `Analytics::read` ay pumipili ng `management()` dahil Admin ang role.
3. Ang card ay galing sa `self::outstanding($p)`: kabuuan ng `billings.net_amount` (status Issued o Partially Paid) bawas ang kabuuan ng `collections.amount` ng mga invoice na iyon.
4. Ang card ay may `page: 'billing'`, kaya kapag pinindot ay pupunta sa Billing & Collections.

9. Formulas at Formula Settings

Lahat ng computation ay dumadaan sa `backend/Core/Settings.php` (formula engine), para iisang lugar lang ang babaguhin kapag nagbago ang formula ng kumpanya.

A. Saan galing ang value

1. **Default:** `config/formulas.php`. Dalawang bahagi: `values` (isang numero bawat setting) at `items` (mga listahan: fund categories, P&L expenses, payroll earnings, payroll deductions). May `source` ang bawat isa (halimbawa "Payroll N14").
2. **Binago sa screen:** kapag nag-Save sa Formula Settings, nasa `app_settings` (values) o `formula_items` (listahan). Ito ang mas nauuna kaysa sa default.
3. **Reset:** binubura ang record sa database, kaya babalik sa default ng file.
4. **Lumang record:** hindi nagbabago. Ang fund request ay may `breakdown_json`; ang payroll ay may `formula_snapshot` (galing sa `Settings::snapshot`).

B. Mga uri ng computation (`calc`)

calc	Formula	Halimbawa
`fixed`	parehong halaga	PhilHealth 75
`percent_of_rate`	% × rate ng biyahe	Misc. Logistics 3%
`per_day`	(amount ÷ divisor) × araw na Present	SSS (775 ÷ 24) × araw
`per_km`	amount × km	
`per_trip`	amount × bilang ng trips sa period	
`fuel`	(km ÷ km per liter) × presyo ng diesel	Fuel & Oil
`route_toll`	RFID/toll na naka-save sa route	RFID
`manual`	tina-type ng tao	Trade Test, SSS Loan
`ledger`	hulog mula sa balance ng empleyado	AR EMP - CA, Uniform, Damages

Ang `only_if_attended` ay katumbas ng `IF(D<>"", ...)` sa Excel: ibabawas lang kung may araw na Present.

C. Payroll line (`Payroll::computeLine`), galing sa DRIVERS PAYROLL.xlsx

```
Araw na Present     = bilang ng attendance na Present sa period (o override)
Regular wage        = daily_rate × araw na Present
Trip pay            = kabuuan ng trip pay rate ng bawat Completed trip na may
waybill
                      (rate ng helper kung helper siya sa trip; route muna,
tapos destination)
Gross pay           = regular wage + trip pay + additional earnings
Deductions          = SSS + PhilHealth + Pag-IBIG + loans at manual
                      + hulog ng Cash Advance, Uniform, Damages (hindi
lalampas sa balance)
                      + Savings / Cash Bond 500 (kung may pasok)
Net pay             = gross pay − deductions
```

D. Projected P&L bawat trip (`Settings::projectedPnl`), galing sa Billing P&L workbook

```
Gross sales   = rate ng client
Fuel & Oil    = (km ÷ 4 km/L) × ₱100
RFID          = toll ng route
Misc.         = 3% (Logistics) at 5% (PDI) ng rate
Salaries      = Admin 769.23, Logistics 1,500, PDI 2,307.69 (dummy values sa
Excel, papalitan)
Net income    = gross sales − total expense
Margin %      = net income ÷ gross sales × 100
```

Nakikita ito sa Clients, Routes & Rates at sa halimbawa ng Formula Settings.

E. Patunay na tugma sa Excel (`tests/formula_test.php`)

Check	Inaasahan (Excel)
Fuel, 143 km	3,575
P&L total expense (rate 15,000, 143 km, RFID 651)	10,002.92
P&L net income	4,997.08
Margin	33.31%
SSS, 6 araw	193.75
PhilHealth, Pag-IBIG, Cash Bond (may pasok)	75, 50, 500
Walang pasok	0 na statutory deductions
Fund categories	100, 100, 200, 300, 200

Para sa mas detalyadong gabay sa pagbabago ng formula, tingnan ang `docs/GABAY_PAGBAGO_NG_FORMULA.md`.

10. Security, roles at module isolation

A. Login at session

Proteksyon	Paano ginawa
Password	`password_hash()` at `password_verify()` ng PHP. Hindi nakikita ang password kahit ng Admin. Hindi bababa sa 8 characters, may letra at numero.
Unang login	`must_change_password = 1` sa bagong account at sa reset. Walang ibang action na papayagan hangga't hindi napapalitan.
Maling password	5 maling subok sa loob ng 15 minuto bawat browser session ay haharangin.
Session	Cookie na `HttpOnly`, `SameSite=Strict`, `Secure` kapag HTTPS, at strict mode. Bagong session id pagka-login (`session_regenerate_id`).
Idle timeout	60 minuto na walang galaw (nababago sa `SESSION_IDLE_MINUTES` sa `.env`).
Deactivated user	Bawat request ay tumitingin kung `active = 1` pa ang user.

B. Mga request

- **CSRF:** lahat ng POST ay may token (`X-CSRF-Token` header o `csrf` field) na sinusuri gamit ang `hash_equals`.
- **SQL injection:** lahat ng query ay prepared statements (`ATTR_EMULATE_PREPARES = false`). Ang mga field helper ay tumitingin ng haba, uri, petsa at pinapayagang values.
- **XSS:** lahat ng text mula sa database ay dumadaan sa `UI.esc()` bago ipakita.
- **CSV injection:** ang value na nagsisimula sa `=`, `+`, `-` o `@` ay nilalagyan ng `'` sa CSV.
- **Headers:** `X-Content-Type-Options: nosniff`, `X-Frame-Options: DENY`, `Referrer-Policy: same-origin`, at `noindex` para sa search engines.
- **Errors:** hindi ipinapakita sa user ang detalye ng database o PHP error; nasa server error log lang.

C. Mga file at folder

- `public/` lang ang dapat bukas sa browser. Ang `backend`, `config`, `database`, `docs`, `scripts`, `storage` at `tests` ay hinaharang ng `.htaccess` (sa root at sa bawat folder: `Require all denied`).
- Ang `.env`, `.sql` at `.log` ay hinaharang din.
- Uploads: sinusuri ang tunay na file type gamit ang `finfo`, random na pangalan, at nasa `storage/private`. Download ay dumadaan lang sa API na may role check.
- Backup download: Admin lang, at ang pangalan ng file ay dapat tugma sa `mulawin_backup_YYYYMMDD_HHMMSS.sql`.

D. Roles: dalawang beses sinusuri

1. **Pahina:** `Access::canOpen()` bago ibigay ang data ng pahina. Ang sidebar ay galing sa parehong listahan, kaya hindi lumalabas ang pahinang bawal.
2. **Action:** sa loob ng bawat action, `requireRole()` o `must(hasRole(...))`. Kahit may tumawag ng API nang direkta, haharangin pa rin.
3. **Dispatcher team:** Car Carrier at Container / Wing Van ay hiwalay. Ang `mustBeTripDispatcher()` ang tumitingin na sa sariling team ang trip (Day at Night shift ay iisang team).
4. **Audit:** read-only. Walang action na pinapayagan sa Audit maliban sa sariling password at notifications.

E. Module isolation

Antas	Kapag may sira	Epekto
Backend module	`Actions::load()` ay nasa try/catch (kasama ang ParseError)	503 sa module na iyon lang; gumagana ang iba
Bahagi ng pahina	`readSections()`	Walang laman ang bahaging iyon at may babala; tuloy ang iba
Dashboard	try/catch sa bawat dashboard	"Some dashboard numbers could not be loaded."
Notification	try/catch sa `Notify`	Hindi napipigil ang pangunahing save

Antas	Kapag may sira	Epekto
Screen file	Hiwalay na `<script>` bawat department file	Ang pahina lang ang may "did not load" na mensahe

Nasubukan ito: sinadyang sirain ang `Accounting/Payroll.php`, at ang payroll ay 503 habang ang trips, billing at dashboard ay 200.

F. Tamang data kahit sabay-sabay

- Lahat ng save ay nasa `transaction()`: buo o wala.
- Ang mga record na binabago ay naka-lock (`SELECT ... FOR UPDATE`), halimbawa ang trip, fund request, truck at payroll period, para hindi magbanggaan ang dalawang user.

11. Deployment, backup, testing at pag-maintain ng code

A. Kailangan ng server

- PHP 8.2 o mas bago, na may `pdo_mysql`, `fileinfo` at `mbstring`.
- MySQL o MariaDB, Apache na may `.htaccess` (`mod_rewrite` para sa folder blocking).
- Writable na `storage/private` at `storage/backups`.

B. Install (local o Hostinger)

1. Gumawa ng bagong empty database (`utf8mb4_unicode_ci`).
2. I-import ang `database/schema.sql`, tapos `database/extension.sql`.
3. Kopyahin ang `.env.example` bilang `.env` at ilagay ang DB values. Sa live: `APP_ENV=production`.
4. Gumawa ng unang Admin: `php scripts/create_user.php admin <username> "<Pangalan>"`, o `public/install.php` gamit ang `INSTALL_TOKEN` (32+ characters), tapos burahin ang token at ang `install.php`.
5. Patakbuhin ang `php scripts/preflight.php`. Dapat lahat PASS.
6. Sa Hostinger: mas maganda kung ang document root ay `RP_Mulawin_FMS_V2/public`. Kung hindi pwede, pwede sa loob ng `public_html` dahil hinaharang ng `.htaccess` ang private folders. Gumamit ng HTTPS.

C. Mga setting sa `.env`

Key	Gamit	Default
`APP_ENV`	development o production	development
`APP_TIMEZONE`	Timezone ng PHP at MySQL	Asia/Manila
`SESSION_NAME`, `SESSION_IDLE_MINUTES`	Pangalan ng cookie at idle logout	rp_mulawin_fms_v2, 60
`DB_HOST`, `DB_PORT`, `DB_NAME`, `DB_USER`, `DB_PASS`	Database	127.0.0.1, 3306, mulawin_v2, root, (blank)
`COMPANY_NAME`, `COMPANY_ADDRESS`	Nakalagay sa payslips at reports	R P MULAWIN TRUCKING SERVICE
`INSTALL_TOKEN`	Para lang sa unang Admin sa browser	(blank)

Huwag i-upload sa GitHub ang `.env` (nasa `.gitignore`).

D. Backup at restore

- **Backup:** Admin > Backup > Create backup. Ang `Admin::backupCreate` ay nagsusulat ng bawat table bilang SQL sa `storage/backups/mulawin_backup_YYYYMMDD_HHMMSS.sql`. Kopyahin din ang `storage/private` (uploads).
- **Restore:** bagong empty database → Import ng .sql → palitan ang `DB_NAME` sa `.env` → ibalik ang `storage/private`.

E. Testing

Test	Ano ang sinusuri	Paano patakbuhin
`scripts/preflight.php`	PHP version at extensions, `.env`, writable folders, bawat module ay naglo-load, mga table	`php scripts/preflight.php`

Test	Ano ang sinusuri	Paano patakbuhin
`tests/formula_test.php`	15 checks laban sa Excel (P&L, fund categories, payroll)	`php tests/formula_test.php` (sa bagong database)
`tests/e2e_workflow.py`	Buong workflow sa API, lahat ng 10 roles, Scenario A, B at C	`python3 tests/e2e_workflow.py` <temporary password ng admin1> (sa empty na TEST database lang)

Ang e2e test ay kailangan ng Python `requests`, tumatakbong app sa `FMS_URL`, at ng `mysql` command line.

Huling resulta (nasa `docs/TEST_REPORT.md`): preflight PASS, formula 15/15, e2e 191 checks at 0 failures, at bawat pahina ng bawat role sa browser ay walang error.

F. Paano magdagdag o magbago

- **Bagong action sa module:** magdagdag ng `'pangalan' => self::pangalan($u, $d)` sa `match` ng `action()`, gumawa ng private function na may `requireRole`, field helpers, `must()` at `transaction()`, at tawagin sa screen gamit ang `App.api('Module.pangalan', data)`.
- **Bagong pahina:** idagdag sa `Access::pages()` (label, icon, section, module, JS file, roles), magdagdag ng branch sa `read()` ng module, at `App.page('id', {...})` sa JS file ng department.
- **Bagong module:** idagdag sa `Actions::MODULES` kasama ang department, gumawa ng `backend/Modules/<Department>/<Name>.php`, at kung may bagong screen file, idagdag sa `$jsModules` ng `public/index.php`.
- **Bagong formula value:** idagdag sa `config/formulas.php` (may group, label, unit at source), tapos gamitin sa code gamit ang `Settings::value('key')`.
- **Bagong table:** idagdag sa `database/extension.sql` (at gumawa ng hiwalay na migration SQL para sa live database).

G. Mga bukas pang tanong sa kumpanya

Ang mga patakarang wala sa PDF ay may working default na galing sa repo o Excel, may `TODO PENDING COMPANY CONFIRMATION` sa code, at nakalista sa `docs/PENDING_COMPANY_CONFIRMATION.md` (halimbawa C01 auto-assign priority, C05

budget periods, C07 fund sa redispatch, C10 tax sa billing). Karamihan ay nababago na sa Formula Settings.